

THE NEW SCHOOL • EUGENE LANG COLLEGE OF LIBERAL ARTS

Homework 01: Patterns and Accumulation

LMTH 2040 — Calculus: A Multidisciplinary Approach

Fall 2026

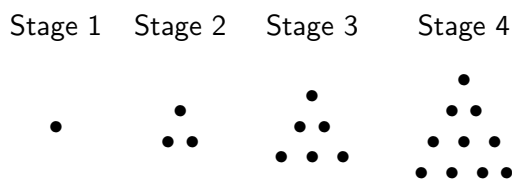
Due Wednesday, September 2

Key Idea

This assignment asks you to investigate patterns before being given formulas for them. Draw pictures, calculate examples, experiment with technology, make conjectures, and explain why you think your conclusions are true.

Problem 1 Triangular Numbers

Consider the following arrangements of dots:



The number of dots in the n th arrangement is called the n th *triangular number*, which we will denote by T_n .

- Draw the next three stages and determine T_5 , T_6 , and T_7 .
- Find a way to arrange, duplicate, reflect, cut, or otherwise transform one of these pictures so that you can determine T_n without adding

$$1 + 2 + 3 + \cdots + n$$

one term at a time.

Include a picture of your construction.

- Use your construction to find a formula for T_n .
Explain why your formula works for *every* positive integer n , not only for the examples you have drawn.
- Now consider the ratio

$$\frac{T_n}{n^2}.$$

Calculate this ratio for several values of n .

What seems to happen as n becomes very large?

Make a conjecture and then use your formula from part (c) to explain whether your conjecture must be true.

Reflection

Computational Experiment.

Use a computational tool of your choice to extend your investigation beyond what would be convenient to calculate by hand.

You might use Python or Jupyter, a spreadsheet, Desmos, GeoGebra, Mathematica, a calculator, or another computational tool.

Investigate

$$\frac{T_n}{n^2}$$

for increasingly large values of n . You might construct a table, make a graph, write a short program, or develop another useful representation.

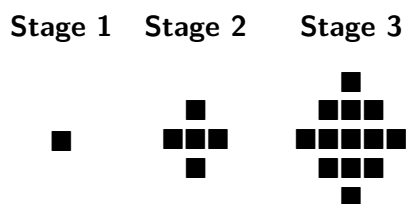
Include enough information that someone else could reproduce your experiment.

Briefly explain:

1. what tool you used and what you asked it to do;
2. what pattern you observed; and
3. what the computation shows—and what it does *not* establish by itself.

Problem 2 A Growing Shape

Consider the following pattern made from unit squares.



Let S_n denote the number of unit squares in Stage n .

- a. Draw Stages 4 and 5.

Determine

$$S_1, \quad S_2, \quad S_3, \quad S_4, \quad S_5.$$

- b. Instead of looking only at the total number of squares, investigate how the figure *grows*.

How many new squares are added in passing from one stage to the next?

Describe the pattern you find.

- c. Find a formula for S_n .

Your explanation should make use of the geometry of the figure rather than simply presenting an algebraic formula.

- d. Compare the sequences T_n and S_n .

What is similar about the way they grow?

What is different?

Can one of the sequences help you understand the other?

Reflection

Computational Experiment.

Use a computational tool of your choice to investigate the pattern at scales that would be inconvenient to draw by hand.

Your goal is *not* simply to generate more terms. Use computation to discover something about the pattern.

For example, you might investigate:

- the total number of squares S_n ;
- the number of new squares added at each stage;
- ratios involving S_n ;
- comparisons between S_n , T_n , n , and n^2 ;
- how quickly different measurements grow;
- or another pattern that you notice.

Present one computational observation that you find mathematically interesting.

State a conjecture suggested by your experiment. Then explain what additional mathematical reasoning would be necessary to establish that conjecture.

Reading Otto Toeplitz

Read the **Preface to the English Edition** and the opening section of Part I at least through the second paragraph on page 6 of “**The Nature of the Infinite Process**,” from Otto Toeplitz, *The Calculus: A Genetic Approach*.

Historical / Cultural Note

Toeplitz describes this book as an attempt to develop a *genetic approach* to calculus. As you read, pay attention to what he thinks is unsatisfactory about the usual presentation of mathematics and what he proposes doing differently.

Response

Write approximately one paragraph responding to the following:

What does Toeplitz mean by a “genetic approach” to calculus?

Identify one claim he makes about how mathematics should be learned or presented. Explain the claim in your own words and then respond to it: what do you find convincing, questionable, or unresolved?

Where useful, draw on your experience with one of the mathematical problems in this assignment.

Submission

Submit a single PDF containing your mathematical work, figures, computational experiment, and reading response.

Your solutions should communicate your reasoning, not only your final answers. Include diagrams when they are part of your argument.

If you use computational tools, briefly identify the tool and describe how you used it. You are responsible for understanding and explaining any mathematics, code, or claims included in your submission.